Add Target Group Block to Your NLB Module

hcl

```hcl
# Add this target group resource to your nlb_api module
resource "aws_lb_target_group" "this" {
  count = length(var.target_groups)

  name        = var.target_groups[count.index].name
  port        = var.target_groups[count.index].port
  protocol    = var.target_groups[count.index].protocol
  vpc_id      = var.vpc_id
  target_type = var.target_groups[count.index].target_type  # "ip" for on-prem

  health_check {
    enabled             = var.target_groups[count.index].health_check.enabled
    healthy_threshold   = var.target_groups[count.index].health_check.healthy_threshold
    interval            = var.target_groups[count.index].health_check.interval
    port                = var.target_groups[count.index].health_check.port
    protocol            = var.target_groups[count.index].health_check.protocol
    timeout             = var.target_groups[count.index].health_check.timeout
    unhealthy_threshold = var.target_groups[count.index].health_check.unhealthy_threshold
  }

  tags = var.tags
}

# Target group attachments (for registering targets)
resource "aws_lb_target_group_attachment" "this" {
  count = length(local.target_attachments)

  target_group_arn = aws_lb_target_group.this[local.target_attachments[count.index].tg_index
  target_id        = local.target_attachments[count.index].target_id
  port             = local.target_attachments[count.index].port
}

# Local to flatten target attachments
locals {
  target_attachments = flatten([
    for tg_index, tg in var.target_groups : [
      for target in tg.targets : {
        tg_index  = tg_index
        target_id = target.id
        port      = target.port
      }
    ]
```

```
  ])
}
```

Update Your Module Variables

Add these variables to your `variables.tf`:

```hcl
variable "target_groups" {
  description = "List of target groups for the NLB"
  type = list(object({
    name        = string
    port        = number
    protocol    = string
    target_type = string
    health_check = object({
      enabled             = bool
      healthy_threshold   = number
      interval            = number
      port                = string
      protocol            = string
      timeout             = number
      unhealthy_threshold = number
    })
    targets = list(object({
      id   = string
      port = number
    }))
  }))
  default = []
}
```

Usage Example (When Calling Your Module)

```hcl
module "nlb_api" {
  source  = "tfe.jpmchase.net/ATLAS-MODULE-REGISTRY/nlb/aws"
  version = "15.0.0"

  name                             = var.neutron_nlb_name
  certificate_arn                  = var.neutron_cert_arn
  enable_cross_zone_load_balancing = true
  on_prem_accessible               = true

  target_groups = [
```

```
    {
      name        = "onprem-api-tg"
      port        = 8802
      protocol    = "TCP"
      target_type = "ip"  # Key for on-premises targets

      health_check = {
        enabled             = true
        healthy_threshold   = 3
        interval            = 30
        port                = "traffic-port"
        protocol            = "TCP"
        timeout             = null
        unhealthy_threshold = 3
      }

      targets = [
        {
          id   = "10.1.1.100"  # Your on-prem IP
          port = 8802
        },
        {
          id   = "10.1.1.101"  # Another on-prem IP
          port = 8802
        }
      ]
    }
  ]

  # Your existing configuration...
}
```

This approach allows you to:

1. Define target groups within your NLB module
2. Set `target_type = "ip"` for on-premises targets
3. Register on-premises IP addresses as targets
4. Keep everything modular and reusable